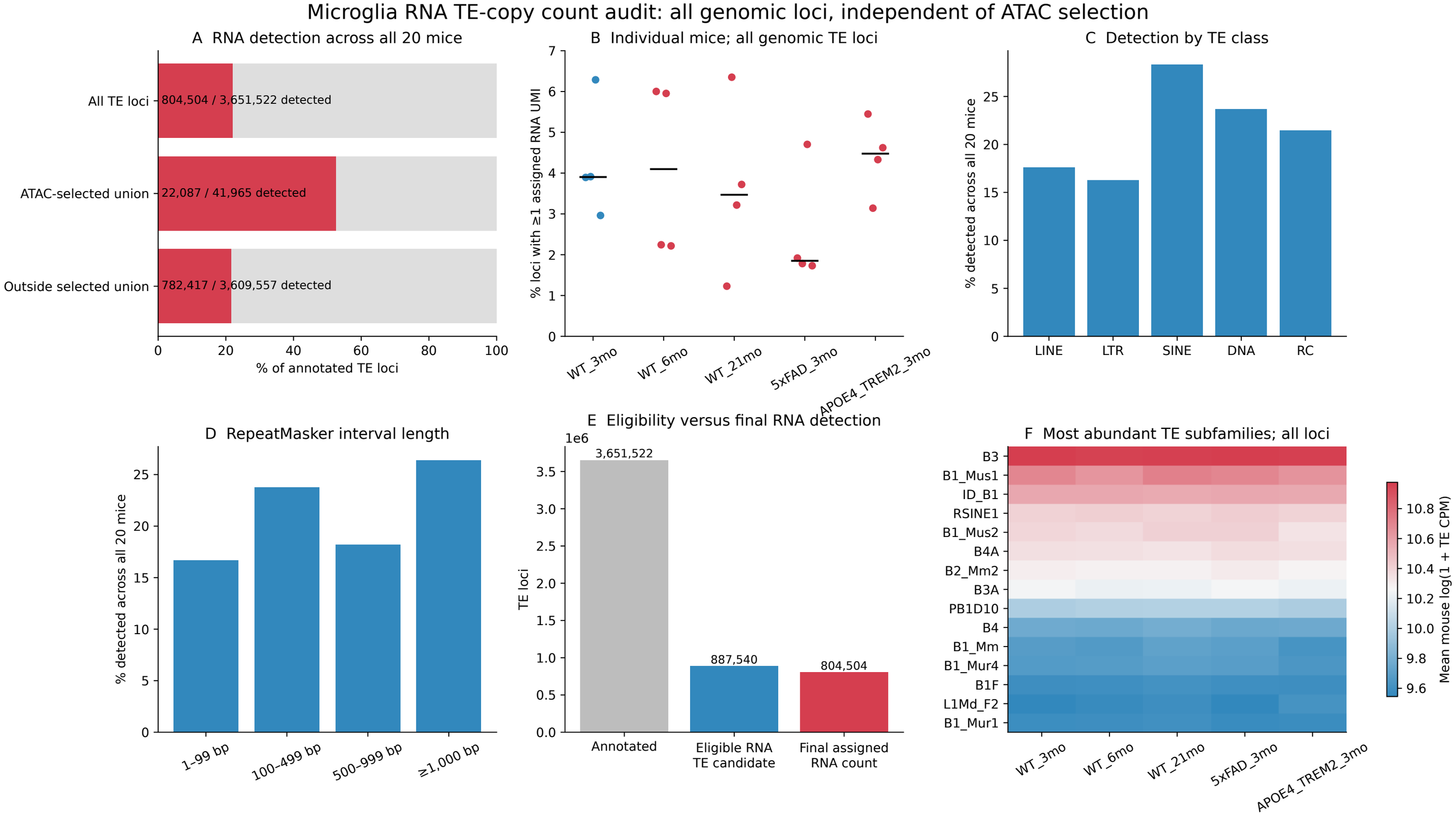
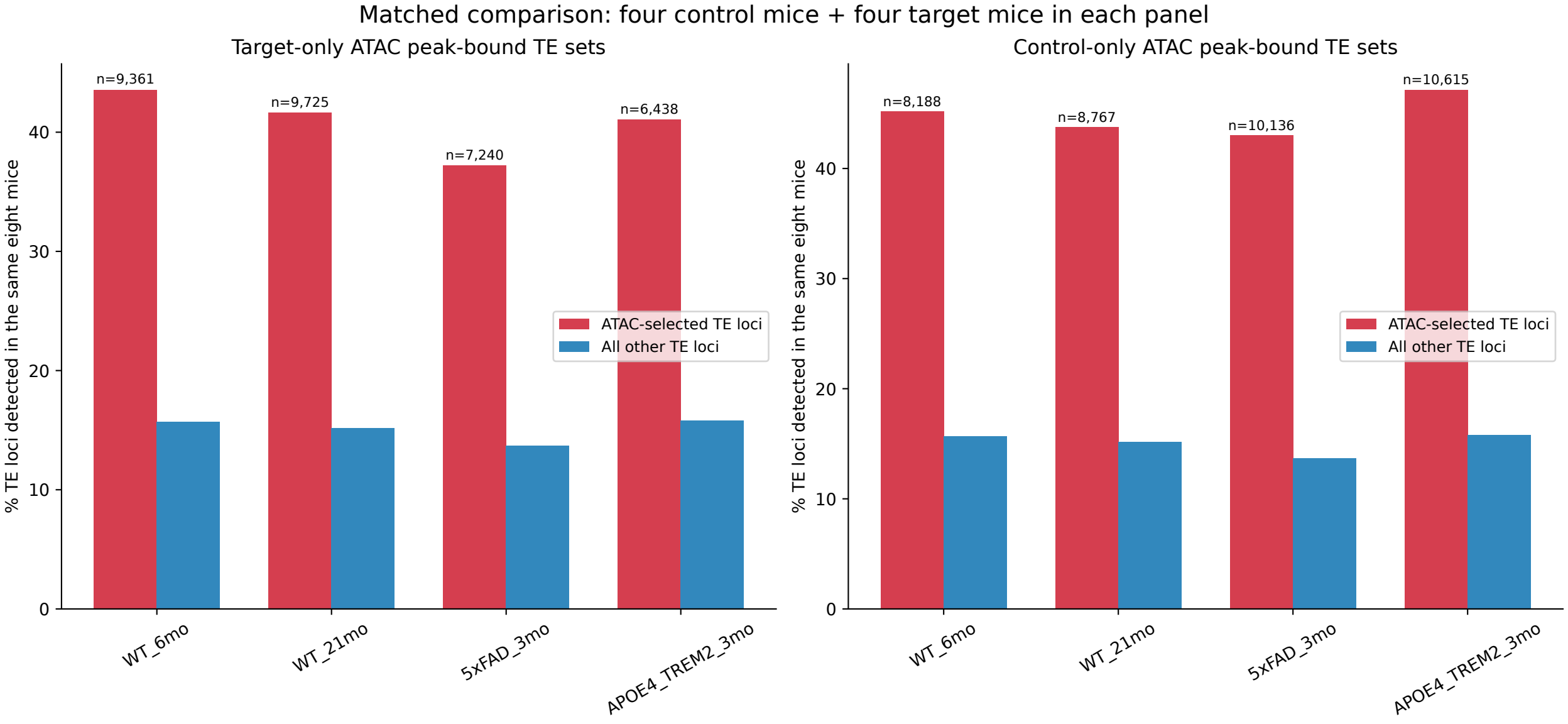

Microglia RNA TE-copy count audit: all genomic loci, independent of ATAC selection
A RNA detection across all 20 mice
All TE loci
804,504 / 3,651,522 detected
ATAC-selected union
22,087 / 41,965 detected
Outside selected union
782,417 / 3,609,557 detected
0
20
40
60
80
100
% of annotated TE loci
B Individual mice; all genomic TE loci
% loci with ≥1 assigned RNA UMI
0
1
2
3
4
5
6
7
WT_3mo
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
C Detection by TE class
% detected across all 20 mice
0
5
10
15
20
25
LINE
LTR
SINE
DNA
RC
D RepeatMasker interval length
% detected across all 20 mice
0
5
10
15
20
25
1–99 bp
100–499 bp
500–999 bp
≥1,000 bp
E Eligibility versus final RNA detection
1e6
TE loci
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
3,651,522
887,540
804,504
Annotated
Eligible RNA TE candidate
Final assigned RNA count
F Most abundant TE subfamilies; all loci
B3
B1_Mus1
ID_B1
RSINE1
B1_Mus2
B4A
B2_Mm2
B3A
PB1D10
B4
B1_Mm
B1_Mur4
B1F
L1Md_F2
B1_Mur1
WT_3mo
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
Mean mouse log(1 + TE CPM)
9.6
9.8
10.0
10.2
10.4
10.6
10.8


Matched comparison: four control mice + four target mice in each panel
Target-only ATAC peak-bound TE sets
Control-only ATAC peak-bound TE sets
% TE loci detected in the same eight mice
ATAC-selected TE loci
All other TE loci
n=9,361
n=9,725
n=7,240
n=6,438
n=8,188
n=8,767
n=10,136
n=10,615
WT_6mo
WT_21mo
5xFAD_3mo
APOE4_TREM2_3mo
0
10
20
30
40